

RAG, Pydantic & Streamlit - Homework Questions

Q1. What is Ragas primarily used for?

- A. Training large language models from scratch
- B. Evaluating the performance of RAG (Retrieval-Augmented Generation) pipelines
- C. Designing website user interfaces
- D. Managing Python virtual environments

Q2. In Ragas, what does the 'faithfulness' metric measure?

- A. How fast the model generates an answer
- B. Whether the generated answer is grounded in the retrieved context (not hallucinated)
- C. The total number of documents stored in the database
- D. The number of parameters in the language model

Q3. What is the main purpose of Pydantic in a Python project?

- A. To create plots and charts
- B. To validate and parse data using Python type hints
- C. To send HTTP requests to a server
- D. To train machine learning models

Q4. To create a Pydantic data model, your class should inherit from:

- A. object
- B. BaseModel
- C. DataClass
- D. PydanticModel

Q5. Given a Pydantic model with a field defined as age: int, what happens if you pass age='twenty'?

- A. It silently sets age to 0
- B. It raises a validation error because 'twenty' cannot be converted to an integer
- C. It stores 'twenty' as a string without any error
- D. It removes the age field automatically

Q6. How do you give a Pydantic model field a default value?

- A. name: str = 'Unknown'
- B. name: str default 'Unknown'
- C. default(name, 'Unknown')
- D. name == 'Unknown'

Q7. What is the main difference between BaseModel and BaseSettings?

- A. They are exactly the same with different names
- B. BaseModel is for general data validation; BaseSettings is for loading configuration, often from environment variables
- C. BaseSettings cannot validate any data
- D. BaseModel can only store text strings

Q8. BaseSettings most commonly reads configuration values (such as API keys) from:

- A. The computer's printer
- B. Environment variables and .env files
- C. The GPU memory
- D. A random number generator

Q9. Which Streamlit function is used to take text input from the user?

- A. st.print()
- B. st.text_input()
- C. st.read()
- D. st.input_box()

Q10. In a Streamlit RAG app, what is the correct general order of steps?

- A. Display answer, then retrieve documents, then user asks question
- B. User asks question, then retrieve relevant documents, then generate answer with the LLM, then display answer
- C. Generate answer, then user asks question, then retrieve documents
- D. Retrieve documents, then display answer, then user asks question